

SLEEVEDRIFT · CALCULATION

SleeveDrift sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First issue for TRL 3 on the constructable design (SVD-DDR-002); fit, hood, haul forces, overload limit, output, casualty, air, set-up, durability, packages, cost

SleeveDrift sizing calculations

On paper SleeveDrift does what it was drawn for except move spoil fast enough. Every piece passes along a 780 mm pipe bore, the heaviest weighs 31 kg, the hood carries 5 kN at 5.5 MPa with under 0.1 mm of deflection, the portal crew turns the winch with 92 N at most, the rope keeps a factor of 6.5 on the 2.5 kN breakaway load, the blower delivers about 15 m³/min at the face, a casualty comes out in about 3.5 minutes, and the kit is set up in about 1.6 hours. The miss is output: one person cranking at a sustainable 60 W moves the loaded train at 3.9 m/min, so a 120 L train takes 32.7 minutes a round trip and the haul carries about 0.22 m³ an hour against R3's 1 m³. Dragging sledges 60 m is expensive in energy, and hand power sets the limit; gearing does not change it. Rope wear over a 72-hour shift (R9) cannot be settled on paper. Every number below is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C3], is the line of the script's output that carries it.

Safety: SleeveDrift is rescue equipment for people working inside a pipe under collapse debris, with ropes under tension and a hand-wound haul. These are first-principles estimates for a paper proof of concept; they do not show that any part is safe. The hood, sheave bracket, ring fixing and anchors must be load tested before use (SVD-BLD-001, section 6), the breakaway swivels must never be bypassed, and nobody is on a rope line or near the sheave while the train moves. See SVD-PRC-001, Safety.

■ Scope and method

The note checks every requirement in SVD-REQ-001 v0.3 against the design in SVD-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and `masses()`, so the ring, sledges and trestles used here are the ones in the STEP files and drawings SVD-DWG-001 and 002. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`, and writes `docs/04-calcs/results.csv`.

The design case is a 60 m rescue pipe of 800 mm outside diameter and 10 mm wall, horizontal, with the portal haul station 5.5 m behind the pipe mouth and the train working at the face.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Pipe	800 x 10 mm steel, 780 mm bore, 60 m, horizontal	Silkyara escape pipe; confirm at the fit trial
Spoil	Loose density 1.7 kg/L (range 1.5 to 1.9); bulking 1.3	Handbook ranges for broken rock, soil and rubble
Sliding	UHMW-PE runners on a gritty steel floor: friction 0.30 sliding, 0.40 starting; rope on the floor 0.50	Conservative for dry grit; confirm in trial
Winches	Two-speed self-tailing, power ratios 13 and 40 with a 250 mm handle, efficiency 0.85	Typical size 40 class; confirm with the winch bought
People	One person cranking sustains 60 W; handle at 60 rpm light; hand-over-hand hauling 0.4 m/s	Ergonomics ranges for shift work
Crew times	Tipping a sledge 1.0 min (two people); loading 30 L/min; signals 1.0 min a cycle	Estimates; measure at TRL 4
Return tension	50 N held on the easing winch	Enough to keep the return leg from snagging
Steel	S355, E = 205 GPa	
Fixing	Crown screw preload 15 kN, side screw 10 kN, friction 0.20 at the pads and invert	M20 class 8.8 at about 60 N m; rusty or painted pipe
Air	200 mm layflat, Darcy factor 0.030, minor losses 2.1 velocity heads, 6.5 % leakage over 65 m	Ventilation duct practice

■ A. Fit in the pipe (R1)

The ring rests on the pipe invert, so the 20 mm diametral clearance is all at the crown [A1]. With the jacking pads screwed back against the shell, every segment fits a 776 mm circle; the sledge is 300 mm wide and 158 mm high; the rolled stretcher is about 480 mm. The largest single piece is the crown segment: chord 658 mm, rise 190 mm, 31.0 kg [A2]. It rides in on a sledge.

With the duct on the upper +Y wall and the return rope on the floor, the largest clear circle left for a crawling person is 554 mm, against 780 mm in the empty pipe [A3]. Inside the ring the clear height is 750 mm [A4]. The 600 mm stretch of R1 has not been designed; the sizes are parameters in the model, but the crawl space in a 600 mm pipe needs its own layout [A5]. R1 is met for the 800 mm pipe and at risk for the 600 mm stretch.

■ B. Hood (R2)

The crown section (120 degrees of 5 mm shell plus the two 50 x 10 mm flanges that run the full length as hood ribs) has an area of 4,953 mm² and a second moment of area of 27.9 x 10⁶ mm⁴ about its own centroid [B1]. Taking all 5 kN on the crown alone, spread over the 500 mm that stands beyond the pipe mouth, as a cantilever from the mouth: M = 1,250 N m, stress 5.5 MPa against 355 MPa yield, tip deflection 0.01 mm [B2]. Across the hood the curved plate acts as an arch: 15.2 kPa gives a hoop force of 5.7 N/mm and 1.1 MPa [B3]. With the head form set 100 mm below the hood, the deflection uses a negligible part of the gap; R2 is met on paper [B4]. A concentrated rock impact is not covered by this check; the CalRig load test settles it.

The moment goes into the pipe through the bottom of the ring at the mouth and the two crown pads 175 mm inside: 3.7 kN on each crown screw, well inside the 15 kN preload [B5].

■ C. Haul forces and the overload limit (R4, R11)

A train of four 29.9 L sledges carries 120 L, 203 kg of spoil, with 33 kg of sledges and links [C1]. 120 m of rope lies in the pipe at any time (the pull leg plus the tail and return legs), dragging 38 N; the back tension on the easing winch adds 53 N through the sheave [C2]. The pull is 786 N loaded, 1,018 N to start and 188 N empty [C3].

Table 2. Handle force at winch A [C4].

Case	High gear (ratio 13)	Low gear (ratio 40)
Loaded, moving	71 N	23 N
Loaded, starting	92 N	30 N

R4 is met in either gear. Without a limit, one person heaving 400 N in low gear would put 13.6 kN on the rope, so a breakaway swivel at each end of the train releases at 2.5 kN (R11) [C5]. The rope (18 kN, 90 % at the splice) then has a factor of 6.5 on the release load and 21 on the working pull [C6].

At the limit the face sheave pin carries 5.0 kN, 35 mm above the base plate: 114 MPa of bending, a factor of 3.1 on S355 yield [C7]. The ring is held against that pull by friction at the pads and the invert: about 79 kN of normal force gives 15.8 kN at a friction of 0.20, a factor of 3.2 [C8].

At the portal, the rope rises 634 mm over 5.5 m to the winch drums: 6.6 degrees, inside the 5 to 10 degrees a self-tailer wants [C9]. The trestle's sling eye is 640 mm up, level with the rope, so the overturning moment at the limit is 25 N m against 119 N m from the trestle's own weight [C10].

■ D. Output (R3)

By hand the loaded train moves at 3.9 m/min (60 W at the handle) and the empty train at 7.2 m/min (handle at 60 rpm) [D1]. A cycle is 15.4 min out, 4.0 min unloading, 8.3 min back in, 4.0 min loading and 1.0 min of signals: 32.7 min for 120 L, or **0.22 m³/h** [D2]. R3 is not met.

For comparison, the Silkyara hand miners advanced about 10 m in under 24 hours through a 0.8 m bore; bulked, that is about 0.27 m³/h of loose spoil [D3]. The hand haul is slower than that face.

Table 3. Output options (estimates).

Option	What changes	Cycle	Output
A. As designed	Hand cranking only	32.7 min	0.22 m ³ /h [D2]
B. Portal drill drive	A right-angle drill with a winch bit on each winch, 120 rpm at the socket: 14.5 m/min, 190 W at the rope	17.3 min	0.42 m ³ /h [D4]
C. B plus a longer train	Six sledges, two people tipping, spoil pre-bagged at the face; pull 1,135 N	15.3 min	0.71 m ³ /h [D5]

None of the options reaches R3 on paper. Loading and unloading dominate once the haul is powered. The options are set out as a decision for Amish in `docs/REVIEW.md`.

■ E. Casualty evacuation (R6)

A 100 kg manikin on a 9 kg stretcher needs 412 N to move and 519 N to start [E1]. Three people hauling hand over hand on the pull rope through winch A (the drum turns freely in the hauling direction and holds when let go) pull 173 N each at the start, at 24 m/min: 3.5 min for 60 m including a minute to clip on [E2]. Cranked in high gear at 60 rpm it would take 8.3 min, so hand hauling is the method [E3]. R6 is met on paper.

■ F. Air at the face (R5)

65 m of 200 mm layflat at 0.27 m³/s runs at 8.6 m/s and loses 525 Pa; the blower needs about 284 W of shaft power [F1]. After 6.5 % leakage, about 15.1 m³/min reaches the face, 5.0 m³/min a person; R5 is met with a wide margin [F2]. The pipe holds 28.7 m³ of air, changed every 1.9 min, and the return air moves back along the pipe at 0.57 m/s [F3]. The body heat of three people at 300 W warms it by 3.0 K [F4]. Gas monitoring is still required (R12); fresh air does not remove gas that seeps in through the debris.

■ G. Set-up time (R7)

Table 4. Set-up sequence on the critical path [G1].

Step	Minutes
Trestles, winches, slings and bins at the portal	20
Blower and first duct lengths at the portal	10
Bottom segment taken in on a sledge by a crew member, both ropes paying out	15
Bottom segment set in the mouth; tail rope reeved round the sheave	10
Crown and side segments sent in on one shuttle trip	8
Segments bolted, jacking screws tightened	20
Train made up and clipped in; first empty run in	8
Signal check and gas check before digging	5

The total is 96 min, about 1.6 h, with the duct and cable hung in parallel by a second pair in about 40 min [G2]. R7 is met on paper; a timed drill verifies it.

■ H. Durability (R9)

At the hand rate, 72 hours is about 132 round trips, and each metre of rope in the pipe drags about 32 km over the floor [H1]. The sledge runners slide about 16 km and lose about 1.6 mm of their 10 mm at an assumed 0.1 mm/km [H2]. Rope abrasion on gritty steel over that distance cannot be predicted on paper, so R9 is at risk [H3]. The sheave, pins and sledges are lightly loaded.

■ I. Packages (R10)

Table 5. Packages, kg [I1].

Package	Mass
Crown segment	31.0
Side segments (2)	33.2
Bottom segment, sheave, spacer and cover	32.1
Winch trestle (each of two)	34.8
Winches (2), handles, ties	22.0
Sledges (4, nested), links, swivels	34.8
Ropes (215 m)	15.0
Blower	20.0
Duct (65 m)	26.0
Stretcher, spreader, monitors, phones, lamps	29.1
Tools, spigot, jacks, bolts, hangers	26.0

The kit is about 304 kg in 12 packages; the heaviest is 34.8 kg, so R10 is met [I2].

■ J. Cost

Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 9,867.50 (USD 4,867.50 over the target) [J1]. The largest lines are the two self-tailing winches (USD 2,300), the four-gas monitors (USD 1,100), the signal line (USD 860), the duct (USD 765) and the blower (USD 750) [J2].

■ Results

Table 6. Results against the requirements (docs/04-calcs/results.csv).

ID	Result	Status
R1	Every piece passes a 776 mm circle in a 780 mm bore; 600 mm variant not designed	Met for 800 mm; at risk for the 600 mm stretch
R2	5.5 MPa, 0.01 mm at 5 kN	Met on paper
R3	0.22 m³/h by hand	Not met
R4	92 N starting in high gear	Met on paper
R5	15.1 m³/min at the face	Met on paper
R6	3.5 min by hand hauling	Met on paper
R7	1.6 h estimated	Met on paper (estimate)
R8	Battery lamps, gas monitors and sound-powered telephones only	Met by design
R9	About 132 round trips; rope abrasion unknown	At risk
R10	Heaviest package 34.8 kg	Met
R11	Release at 2.5 kN; rope factor 6.5	Met by design
R12	Two four-gas monitors in the kit	Met by design